NATIONAL UNIVERSITY

Managerial Statistics
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WRITTEN ANALYSIS OF THE CASE
Final Requirement
Case Title: *Pitcher Perfect: Visualizing Interactive Beer Profiles*

I. CASE SYNOPSIS

Pitcher Perfect Corp.(PP), a craft beer manufacturer known for its playful, pun-filled beer names (*You Make My Heart Malt*, *Ale Yeah!*, *Brewness as Usual*) and unique flavors, is facing a cultural shift. Traditionally, the company relied on the intuition and creativity of its brewers, who viewed their craft as an art form. However, new executives are pushing for a data-driven approach to product development, leading to tension between the old and new ways of thinking.

This case study focuses on DeMario Moore, a new product manager navigating this transition as he develops the company's next beer flavor. Can he balance his creative instincts with the insights gleaned from data analysis? Pitcher Perfect's journey reflects a broader trend in the industry: the integration of data analytics into traditionally creative fields. This case highlights the challenges and opportunities that arise when a company tries to balance data-driven decision-making with artistic expression.

CASE OBJECTIVE

To explore how Pitcher Perfect can utilize data-driven insights to develop a new beer flavor. The focus is on using an Interactive Presentation (IPP) to foster collaboration among employees, encourage innovation, and promote a culture of continuous learning within the company*, but ultimately* gain stakeholders buy-in for the *new beer product*.

Given this objective, the learners will continue to analyze the dataset to identify potential areas where data can enhance, rather than replace, the creative process. We can also look for patterns or insights that might spark new ideas for beer flavors, aligning with Pitcher Perfect's tradition of unique and playful names.

CASE FACTS

Case Characters	Role	Persona	Motivation
DeMario Moore	PP's New Product Manager	Creative **Brewmaster**	• DeMario takes pride in his creative abilities and wants to continue developing successful beer flavors. • While initially resistant, he is motivated to learn and adapt to the company's shift towards data-driven decision-making to remain relevant and valuable. • He seeks recognition for his work, especially from the new CEO, Nathalie Aimé.
Shandi Clark	Vice President of New Product Development	Visionary Product Developer	• Shandi is likely motivated to drive innovation and maintain Pitcher Perfect's reputation for unique and successful beer flavors.
Ana Diaz-Hernandez	Vice President of Marketing	Marketing Maven	• Ana is likely motivated to ensure the successful marketing and promotion of new beer flavors. • She seems supportive of DeMario and encourages him to explore data-driven approaches.
Nathalie Aimé	CEO of Pitcher Perfect	Modernization Champion	• Nathalie is motivated to modernize Pitcher Perfect's operations and decision-making processes by incorporating data science and business analytics. • She aims to establish PP as a leading craft beer brand and likely sees data-driven insights as crucial to achieving this goal.
Himanshu Singh	Senior Data Scientist	Data Science Enthusiast	• He is willing to help other departments, like DeMario's, leverage data for their own goals.
Will Patch	Founder of UVA MSBA (Competitor)	Rival Brewer	• Will is to launch a new craft beer product, Cape Cod'r.

II. CASE ANALYSIS

Pitcher Perfect's Brewmaster DeMario Moore possesses a natural talent for crafting unique beer flavors. Data analytics can amplify his expertise by providing insights into consumer preferences, allowing him to refine his creative vision and validate his intuition. By leveraging data and presenting it in an interactive format (IPP), Moore can gain stakeholder support and foster a culture of collaboration and learning within the company. This approach demonstrates how talent and data can work together to drive innovation and success in the workplace.

With the tools and data provided to DeMario, this now allows him to refine and validate his creative instincts. With years of experience and a deep understanding of his field, he possesses a strong intuition about what beer flavors will resonate with consumers.

The challenge lies in effectively communicating his vision and gaining stakeholder buy-in, as such, the following are the learners' findings and interpretations from the dataset.

BREWMASTER MOORE'S DATA EXPLORATION, SCOPE, SELECTION CRITERIA

While the CEO is interested in the marketability of fruit-flavored beer, Brewmaster DeMoore already has a vision for a new, flavorful beer targeted towards **affluent African-American partygoers in the US**. He understands that the **right amount of hops** is crucial for a great beer.

The dataset from the Beer Profile report contains 3,197 unique beers with a total of 745,812 reviews. However, to prevent skewing the analysis and ensure the focus remains on well-regarded beers, DeMario will now then use probability distribution to classify beer popularity, which aims to avoid the undue influence of highly reviewed (5 rating) but potentially less relevant beers on his analysis of beer characteristics and attributes.

			Range		
Beer Popularity Class	**Probability Distribution**	**Review Bracket**	**Min**	**Max**	**Rank**
Legendary	0.1	329	2962	and Up	1

Epic	0.25	822.5	2140	2961	2
Rising Stars	0.3	987	1153	2139	3
Craft Favorites	0.25	822.5	330	1152	4
Up-and-Coming/Emerging	0.1	329	1	329	5

Moore then filtered the dataset to include only those classified as "Legendary," "Epic," or "Rising Stars." This reduced the sample to 25% of the original dataset, resulting in approximately 183,970 reviews. These three classes are now the considered "*popular-only*" beers.

	Beer Popularity Class	Total Reviews	Unique Beer ID	Reviews per Beer
1	Legendary	6,290	2	3145
2	Epic	33,591	14	2399
3	Rising Stars	144,089	96	1501
4	Craft Favorites	344,643	581	593
5	Up-and-Coming/Emerging	217,199	2504	87
Grand Total		**745,812**	**3,197**	

Now that he was able to narrow the focus to the only relevant or "popular-only" beer, Brewmaster DeMoore now aims to understand the elements of a "great" beer and how they influence consumer perception, especially among affluent African-American partygoers.

DeMario's analysis will focus on flavor, reviewed beer attributes and their relationship with overall ratings. By identifying the specific attributes and characteristics that resonate most strongly with consumers, DeMoore and his audience (thru the IPP session) are now guided in formulating a recipe that delivers a truly exceptional beer experience, one that embodies the spirit of celebration and social connection that he envisions for his target market.

This mini-heatmap is the summary of the learners' findings from the *popular-only* beer dataset. DeMoore can prepare this to illustrate that taste is a strong factor for beer drinkers.

Correlation Matrix
(Expressed in R_2 for Review by Beer Attribute , emphasis on Taste and Overall Review Ratings)

	Overall Ratings	**Taste**	**Appearance**
Taste	86.92%		
Palate	**94.16%**	**98.70%**	
Aroma	77.91%	82.90%	73.21%
Appearance	77.03%	86.87%	

Summary:

The Correlation Matrix emphasizes the importance of taste in predicting a beer's overall rating and suggests that prioritizing taste during brewing can lead to higher consumer satisfaction. The strongest correlation is the gustation factor (taste). Only secondary is olfactory senses (smell), and finally appearance comes least.

Refer to *Appendix A CHARTS AND FINDINGS* for a more detailed analysis of other factors.

Section IV Appendix are scatter plots used for linear regression to analyze the relationship between various beer attributes and the overall rating of the beer in respect to "review_taste".

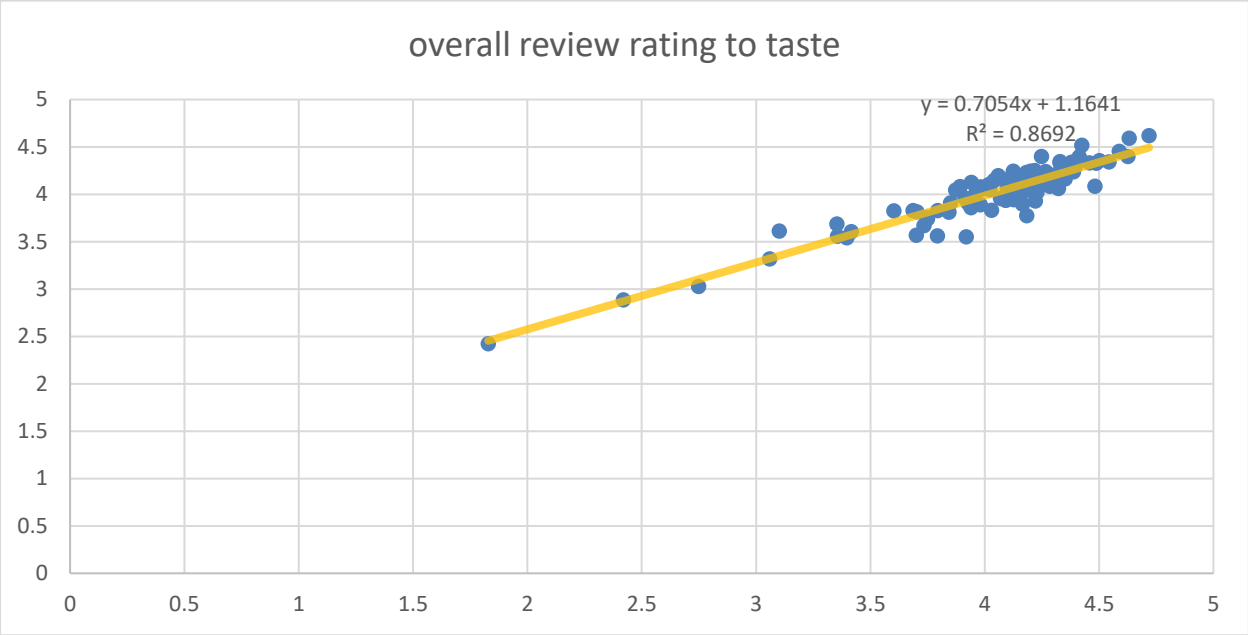
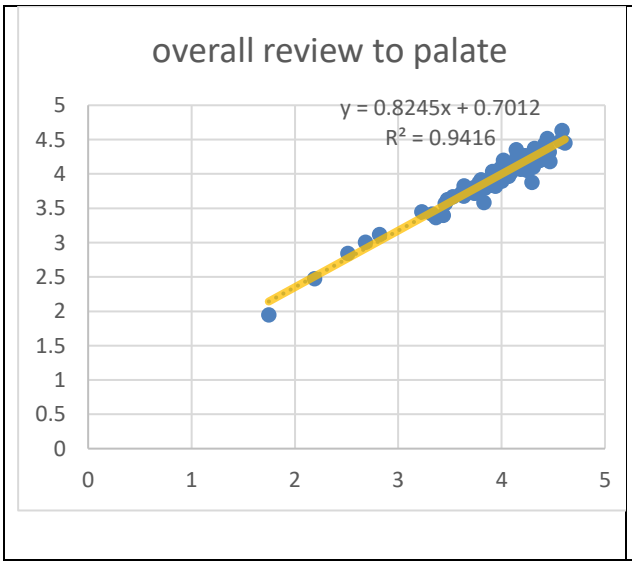
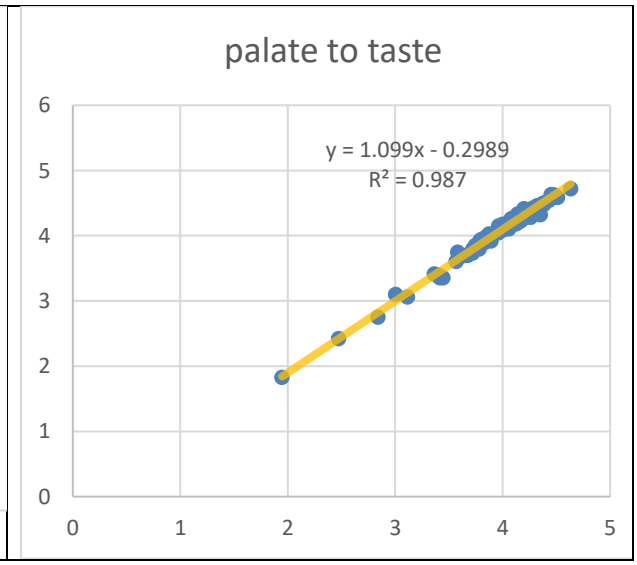


Chart 2-A: "overall_ratings" having a strong correlation with "taste"





Other Findings:

- The learners' linear regression analysis confirms that there is a strong positive relationship between "review_taste" and "review_overall". **This means that taste is a significant predictor of a beer's overall rating.**
- The regression coefficient associated **with "review_taste" indicates how much weight consumers place on taste when forming their overall opinion of a beer**.
- This insight is valuable for developing a new beer product, as it suggests that prioritizing taste during the brewing process is likely to lead to higher overall ratings and greater consumer satisfaction.

QUANTITATIVE METHODS RESULTS

Given that consumer taste is a major factor in a beer's overall rating, DeMoore will identify the top-rated beer using the Bayesian Average method, which considers both taste and the number of reviews for a more reliable assessment.

The top-rated beer identified through this method is **Pliny the Elde**r, an Imperial IPA by Russian River Brewing Company, with an adjusted rating of 4.403803735. (Refer to the learners' detailed derivations in "bayes" sheet of the learners' work file linked here.)

Beer Profie * Link Here	Pliny the Elder is a renowned double IPA known for its balance of bitterness and fruity hop aromas, featuring amarillo, centennial, CTZ, and simcoe hops.
Brewery's History	Brewed by Russian River Brewing Company, it was first created in 1999 by brewmaster Vinnie Cilurzo, who is credited with inventing the double IPA style.
Limited Availability	The beer's scarcity adds to its appeal, with distribution mainly in California and select states. It can occasionally be found online at retailers like Total Wine or Drizly.

Cult Status	It has a perfect score on Beer Advocate and is considered "World-Class," making it a cult favorite among beer enthusiasts. Russian River Brewing Company is also praised for its other beers, which are worth trying.
Official Link	Pliny the Elder | Russian River Brewing Company

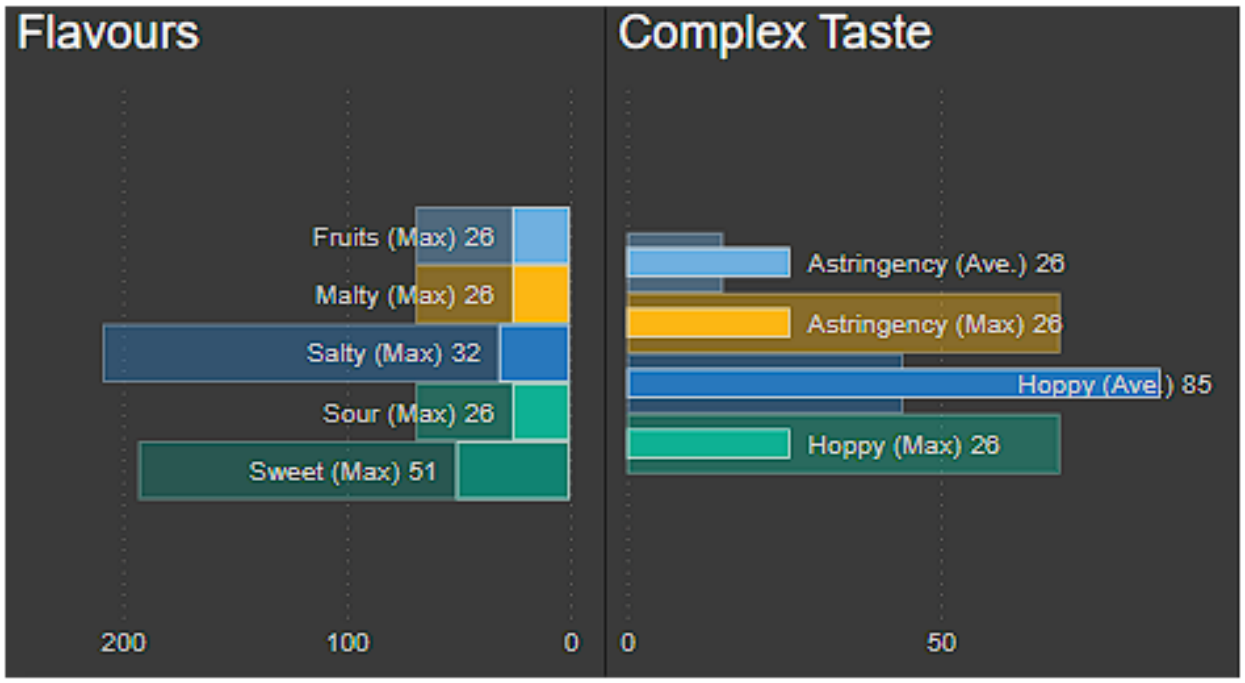
INTRODUCING: PICTHER PERFECT'S INTERACTIVE PRESENTATION (IPP) *

DeMoore will now utilize the Interactive Presentation (IPP) to delve deeper into the characteristics of this beer, exploring its flavor profile and other attributes that contribute to its high rating.

Pliny the Elder is a Double/Imperial IPA brewed by Russian River Brewing Company in Santa Rosa, California. It's considered one of the most sought-after and highly rated beers in the craft beer world.

Beer	Baysean Average Rating (Best Tasting)	Total Reviews (Count)	Overall Rating (Ave.)	Taste (Ave.)
Pliny The Elder	4.40	2527	4.59	4.63
Founders KBS (Kentucky Breakfast Stout)	4.36	1932	4.40	4.63
Trappist Westvleteren 12	4.34	1272	4.62	4.72
Trappistes Rochefort 10	4.33	2170	4.34	4.54
Founders Breakfast Stout	4.32	2502	4.35	4.50
St. Bernardus Abt 12	4.30	2217	4.33	4.49
The Abyss	4.30	1412	4.45	4.59
Bourbon County Brand Stout	4.27	1710	4.08	4.48
La Fin Du Monde	4.26	2483	4.30	4.40
Weihenstephaner Hefeweissbier	4.26	1981	4.52	4.43
Supplication	4.25	1053	4.43	4.57
Tröegs Nugget Nectar	4.25	1955	4.40	4.41
Dreadnaught IPA	4.25	1481	4.33	4.46
Stone Ruination IPA	4.24	2704	4.16	4.35
Pliny The Younger	4.24	610	4.60	4.72
90 Minute IPA	4.24	3290	4.15	4.33
Founders CBS Imperial Stout	4.23	637	4.59	4.70
Schneider Aventinus	4.23	1899	4.23	4.39
Chimay Grande Réserve (Blue)	4.23	2009	4.25	4.37
Duvel	4.22	2450	4.34	4.33
AleSmith Speedway Stout	4.21	1134	4.28	4.44
Hop Rod Rye	4.21	1913	4.22	4.34
Trappistes Rochefort 8	4.21	1561	4.23	4.36
Trappist Westvleteren 8	4.20	707	4.51	4.55
Consecration	4.20	844	4.30	4.47
Furious	4.19	1224	4.34	4.38
Temptation	4.19	847	4.39	4.46
Ten FIDY	4.19	1729	4.17	4.32
Dark Lord Imperial Stout	4.19	1011	4.15	4.41
Alpha King Pale Ale	4.19	1531	4.33	4.33
Parabola	4.18	595	4.42	4.55
Wisconsin Belgian Red	4.18	826	4.24	4.43
Heady Topper	4.18	469	4.63	4.61
Bourbon County Brand Coffee Stout	4.18	456	4.32	4.62

Beer	Baysean Average Rating (Best Tasting)	Total Reviews (Count)	Overall Rating (Ave.)	Taste (Ave.)
Pliny The Elder	4.40	2527	4.59	4.63



IPP (VIA POWERBI) LIVE DEMO HERE

Having identified Pliny the Elder as the top-rated beer and explored its characteristics through the IPP, DeMoore can now leverage this data-driven benchmark to refine his own creative vision for a new beer targeting affluent African-American partygoers who appreciate the celebratory atmosphere akin to that of the hip-hop world.

The interactive nature of the IPP allows for real-time collaboration and exploration with the stakeholders, enabling DeMoore to address any questions or concerns stakeholders might have about the top-rated beer's flavor profile and its potential influence on the new product.

With the data-driven insights in hand, DeMario can confidently circle back to his initial creative concepts, drawing upon his expertise and understanding of consumer preferences to select the most fitting tagline.

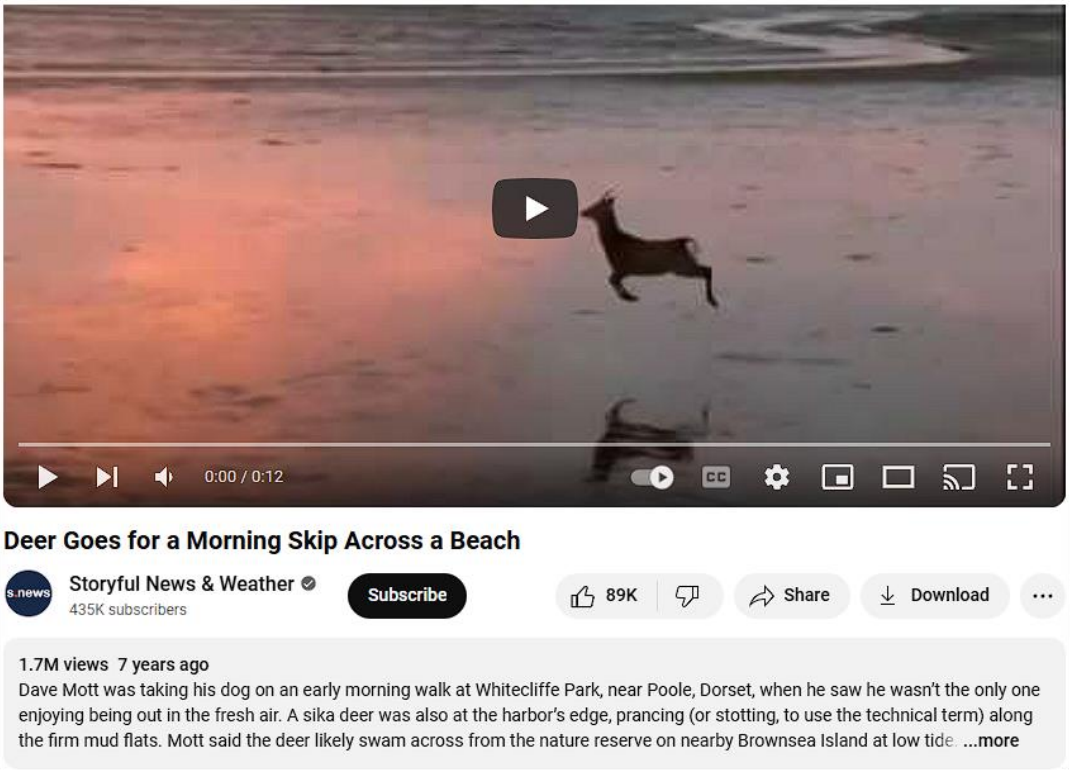
New Beer Candidates DeMario previously conceptualized:

In pursuit of Hoppiness
ReinBeer
Don't worry be HOPPY
This Is How We Brew It?
Let's Get Hoppy
Hop-Town Funk you Up

In this case, DeMario Moore might ultimately choose **"ReinBeer"** as the name for the new tasty beer with the *right amount of Hops* , a playful and festive name that aligns with his vision for a celebratory beverage *that resonates with the target audience*. To further illustrate this vision and generate excitement, DeMario could then direct the audience's attention to a YouTube clip that captures the essence of the ReinBeer concept, subtly hinting at a potential new beer marketing campaign with the tagline--

"**ReinBeer**: *Once You Hop, You Can't Stop.*"

YouTube Link: Deer Goes for a Morning Skip Across a Beach (youtube.com)



III. CONCLUSION

In the world of work, there are those who rely on their innate talents, like Pitcher Perfect's Brewmaster DeMoore with his exceptional sensory abilities. Others depend on learned skills, honed through countless hours of practice. While talent offers a natural advantage, skills can be acquired and refined through dedicated effort.

The tools available at Pitcher Perfect, including data analytics, can amplify DeMoore's innate talent and expertise, allowing him to refine and validate his creative instincts. With years of experience and a deep understanding of his field, DeMoore possesses a strong intuition about what beer flavors will resonate with consumers. The challenge lies in effectively communicating his vision and gaining stakeholder buy-in.

By leveraging data to support his ideas and presenting them in an engaging, interactive format (IPP), DeMoore can bridge the gap between his creative vision and the data-driven priorities of the new executives. This approach not only validates his expertise but also fosters a culture of collaboration and continuous learning within the company. DeMoore's knowledge of the industry and consumer preferences, combined with data-driven insights, positions him to create a successful new beer flavor that aligns with both his creative vision and the company's evolving goals.

In essence, the Brewmaster Moore's approach demonstrates how talent and data can work in harmony to drive innovation and success. By combining his innate abilities and being ***data-informed***, he can create a new beer that not only satisfies his creative vision but also resonates with a broader audience. This IPP approach exemplifies the power of leveraging both talent and tools to achieve optimal outcomes in the workplace.

IV. APENDIX

A. CHARTS AND OTHER FINDINGS

number_of_reviews	
Mean	233.2849546
Standard Error	6.398990495
Median	93
Mode	2
Standard Deviation	361.8118471
Sample Variance	130907.8127
Kurtosis	11.87018712
Skewness	3.029871953
Range	3289
Minimum	1
Maximum	3290
Sum	745812
Count	3197
Confidence Level(95%)	12.54654207

Table 4-A number_of_reviews Descriptive Statistics result

Hoppy-ness Test	
Beer Attribute: ***Hoppy***	
Mean	40.92461683
Standard Error	0.537717625
Median	33
Mode	16
Standard Deviation	30.4036406
Sample Variance	924.3813619
Kurtosis	0.912552084
Skewness	1.10176928
Range	172
Minimum	0
Maximum	172
Sum	130836
Count	3197
Confidence Level(95.0%)	1.054306456

Table 4-B. Hoppy beer attribute - Descriptive Statistics result

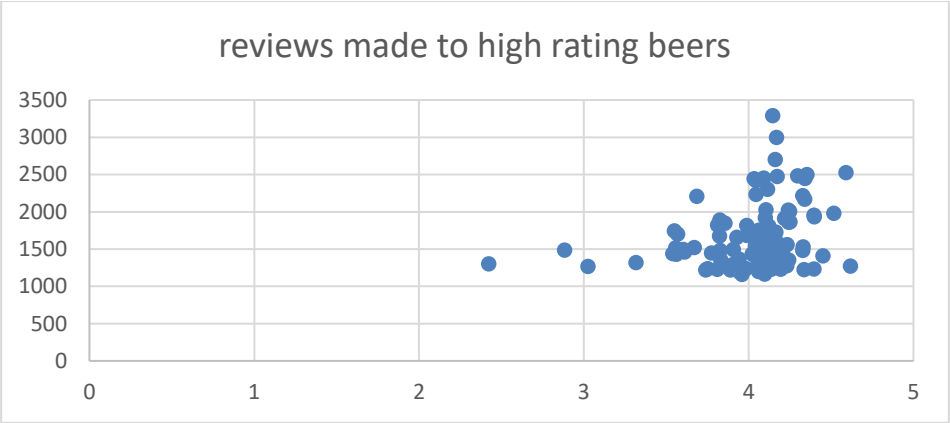


Chart 4-A – High Rating Beers (3, 4, 5) total number of reviews

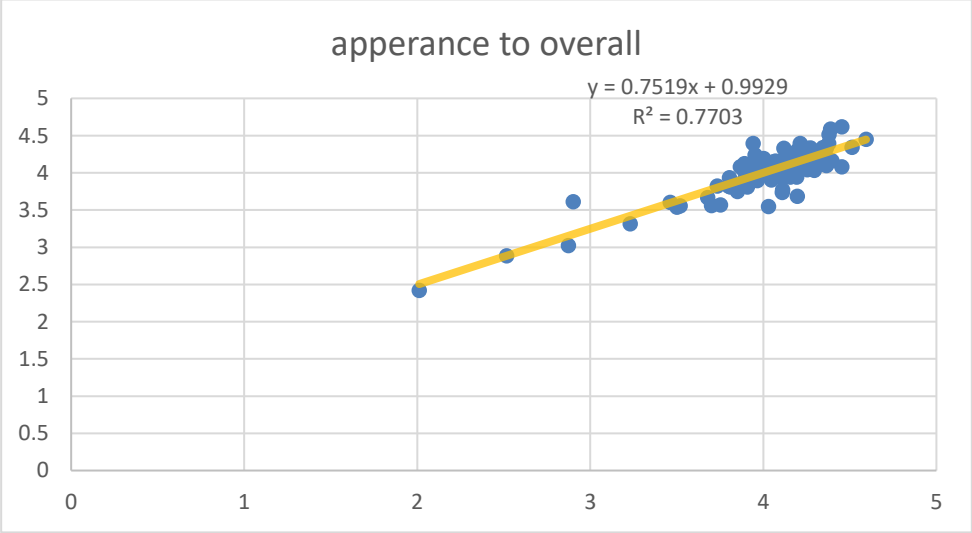


Chart 4-B – review_apperance to review_overall

There are multiple outlier points that are obvious here, this can be studied/ or analyed furthe as it might be an indicator of something like of a 'niche' market.

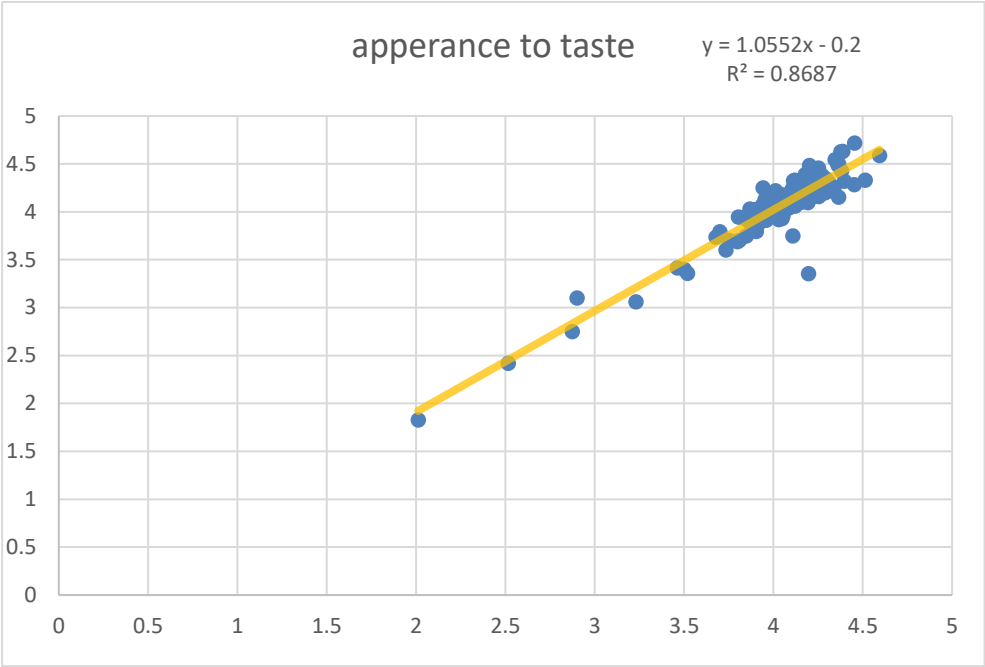


Chart 4-C – review_apperance to review_taste

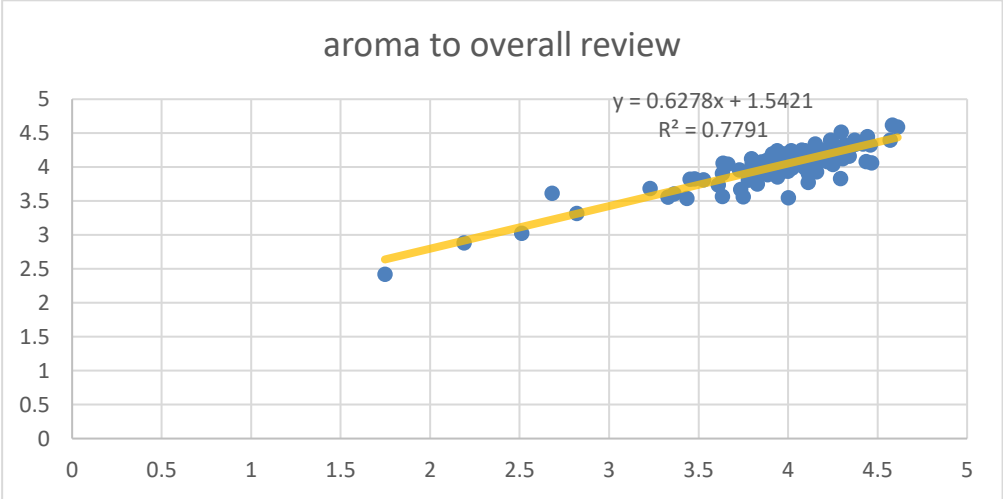


Chart 4-D – review_aroma to review_overall

This indicates a strong relationship between review_aroma and review_overall. This suggests that beers with higher aroma ratings tend to have significantly higher overall ratings. It can be pointed out that a beer's aroma can set expectations for its flavor and influence the overall drinking experience.

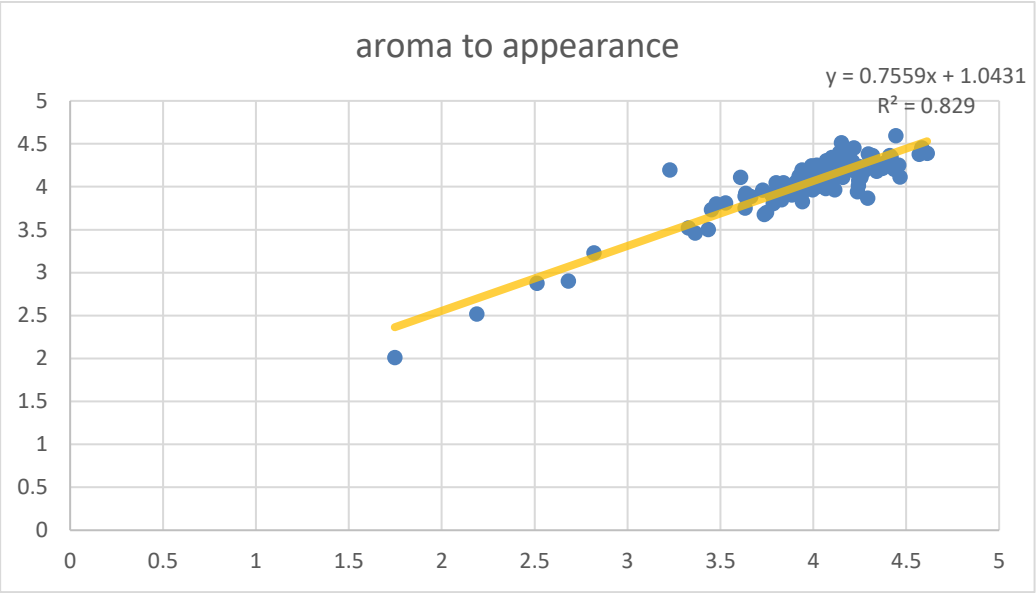
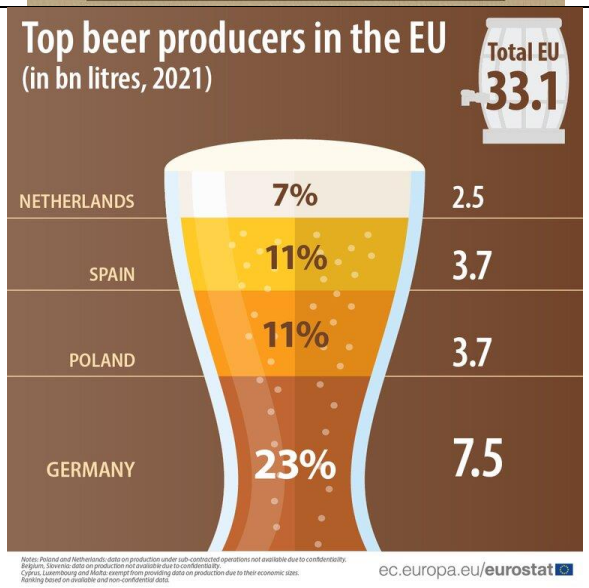
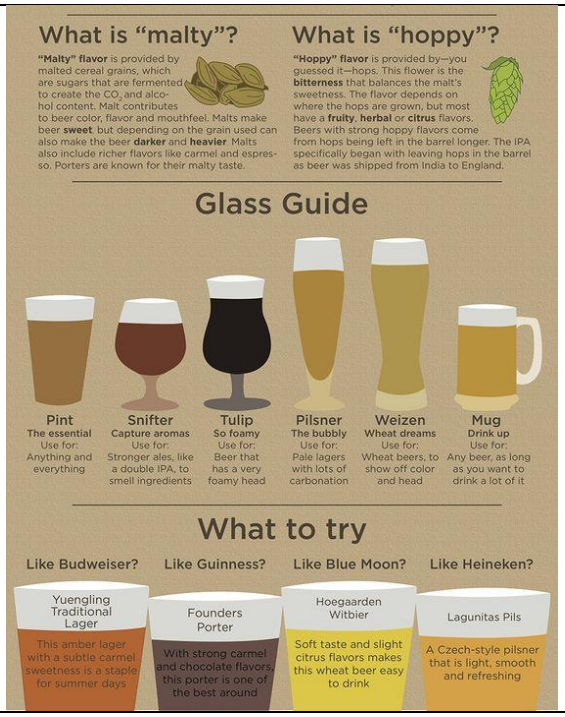


Chart 4-E – review_aroma to review_overall

B. DATASET DATA DICTIONARY

All Dataset Variables definitions are retained from the Case file.
The following are the additional metrics derived by the learners:

1. Beer Popularity Class – Identified possible beer classification by popularity.
2. Beer Popularity Category - 1 being highest, based on Beer Popularity Class
3. Beer Popularity Rank – All researched 3197 beers, ranked by popularity.
4. Hoppy Linear Scaling (1 to 5) – An interval scale representing 5 with the highest hop contents/rating.
5. ABV Linear Scaling - An interval scale representing 5 with the highest Alcohol By Volume (ABV) contents/rating.
6. Baysean Average Rating (BAR) – provides a weighted average of review_overall in respect to taste.

C. KNOW YOUR BEER